Onur Altay Topaloğlu

Computer Vision & ML Engineer · Depth Estimation, Point Tracking & Vision-Language Models

hello@onuraltay.dev | onuraltay.dev | Google Scholar | GitHub | LinkedIn | Munich, Germany

Seeking a **Werkstudent** position in Munich · **20 h/week** · No sponsorship required (student visa) · German B1 · English C1

Profile

M.Sc. Informatics student at TUM with two years of hands-on computer vision work: depth estimation benchmarks, point tracking and optical flow in video, and large-scale evaluation of vision-language models on GPU clusters. Co-first author at a CVPR 2026 workshop. Experienced across the full evaluation loop: curating data, building ground truth, running models at scale on HPC, and turning results into clear findings.

Education

10/2026 – 2028 **Technical University of Munich (TUM)** Munich, Germany
M.Sc. Informatics

09/2021 – 06/2026 **Middle East Technical University (METU)** Ankara, Türkiye
B.S. Computer Engineering (English) — CGPA **3.61/4.00** (≈ 1.6 German scale)
Relevant coursework: Introduction to Deep Learning, Computer Graphics, Cloud Computing

Experience

11/2024 – 03/2026 **ROMER — METU Center for Robotics and AI** Ankara, Türkiye
Undergraduate Researcher, Computer Vision

- **Co-first author** of *LuMon* (CVPR 2026 W), a lunar monocular depth estimation benchmark of 15 models across 6 datasets; wrote its open-source evaluation codebase (depth-vs-disparity handling, per-image affine alignment, dataset-specific masking) and integrated MiDaS v3.1 and Lotus.
- Built the **stereo rectification** and masking stages of the ground-truth depth pipeline for real Chang'e-3 rover imagery (camera calibration + ORB matching); released as 2 public HF datasets.
- Identified a **sim-to-real gap**: synthetic fine-tuning gains (LuSNAR δ_1 0.31 $\rightarrow$ 0.93) did not transfer to real lunar imagery.

09/2025 – 07/2026 **Harvard–MIT & METU Research Collaboration** Remote
Undergraduate Researcher, Trustworthy Multimodal AI

- Owned the fairness study of a construct-alignment framework (AAAI 2027 submission): ran **7 vision-language models** (SmolVLM, Qwen2.5-VL 3B–32B, Gemma-4) on A100 GPUs over demographic counterfactuals on medical (glaucoma, dermatoscopy) and general benchmarks.
- Built an async, resumable **LLM-as-a-judge** harness with 4 judge models and group-fairness metrics; 11K+ VLM inferences and 22K+ judge calls on one benchmark alone.
- Found that scale did not imply fairness: the 32B model flipped 40% of glaucoma diagnoses when only patient age changed, vs. 3% for the 7B model.

06/2025 – 08/2025 **KUIS AI Center, Koç University** Istanbul, Türkiye
Computer Vision Research Intern, Autonomous Vision Group

- Benchmarked 3 SOTA **point trackers** (CoTracker3, Track-On, BootsTAPIR) on SLURM clusters, with shared SIFT seeding and side-by-side visualization for a fair comparison.
- Segmented moving animals in video with RAFT **optical flow** + FlowSAM, after showing that SAM / SAM 2 prompting was insufficient for the task.

06/2024 – 07/2024 **KKB Kredi Kayıt Bürosu** (Türkiye's national credit bureau) Istanbul, Türkiye
AI Intern — built a hybrid dense + keyword retrieval system in PyTorch, outperforming a keyword-only baseline.

09/2024 – 06/2025 **Computer Engineering Department, METU** Ankara, Türkiye
Teaching Assistant — CENG213 Data Structures (C++), CENG232 Logic Design

Publications

CVPR 2026 W *LuMon: A Comprehensive Benchmark and Development Suite with Novel Datasets for Lunar Monocular Depth Estimation.* Co-first author, AI4Space Workshop. [Paper] [Code] [Datasets]

Under review A construct-alignment framework testing whether foundation-model evaluations measure the construct they claim to. Co-author, AAAI 2027.

Projects

2025 – 2026 **Capture & Cook — fridge photo to recipe recommendation** METU
Senior Design Project [Website]

- Compared Gemini and Qwen VLM APIs on a 100-photo test set and shipped Qwen for ingredient detection (more accurate at similar latency, lower cost); built hybrid TF-IDF + FAISS recipe search.

02/2026 – 08/2026 **LawPal — AI-native workspace for lawyers** Istanbul, Türkiye
Co-creator — full-stack AI product built end to end (Next.js, Supabase, Claude API, MCP)

Technical Skills & Languages

Vision depth estimation, stereo rectification, point tracking, optical flow (RAFT), SAM / SAM 2, VLMs, feature matching (SIFT, ORB), diffusion & flow matching

ML & Tools PyTorch, Hugging Face Transformers, OpenCV, NumPy, SciPy, Pandas, FAISS

Eng. & Compute Python, C/C++, TypeScript, SQL · CUDA, Slurm (HPC), Docker, Linux, Git

Languages Turkish (native), English (C1, TOEFL iBT), German (B1)

Interests Calisthenics, boxing, running